

Sreeram_experimental — Branch Overview

BUSI70575 (Imperial) · meta-labelling metamodel · 11 instruments (equity / energy / metals) · 4,886 events · 2026-06-03

Concise overview of what was built, what was measured, and the decisions made — for the submission-readiness review. Not the academic report itself.

1. Data

Source	What	Range	Use
data/ohlc_data.csv	Daily OHLCV for 11 futures	1990 → Jun 2022	labels, F1–F17, F21 features
data/primary_signals.csv	Primary signal {−1, 0, +1}	Jan 2020 → Jun 2022 (645 dates)	trigger, trade direction
data/bloomberg/cleaned/macro_harry.parquet	21 macro series (VIX, rates, credit, FX, 1990 PMJ)	1990 → Jun 2022 daily	F11 macro features
data/bloomberg/cleaned/futures_term.parquet	5 quotes + 2nd month for 8 commodities + 1990 futures	Jan 2020 → Jun 2022 (UX from 2014, XN from 2005)	F18, F19, F20
data/bloomberg/cleaned/options_iv.parquet	Quarter 3m ATM IV + 90 % / 110 % MNY	2005 → Jun 2022	F19 options IV
data/bloomberg/cleaned/eia_crude.parquet	EIA weekly crude inventory change	1990 → Jun 2022 (Friday data)	F22–F25 crude oil

Hidden test is H2-2022; Bloomberg data cannot legally be extended past Jun 2022 (released-period constraint). The model architecture is robust to BBG missingness at hidden-test time — verified by simulated-missingness ablation.

2. Features

105 features registered across 18 families; 80 kept after the drift filter. Drift filter rule: drop a feature if its train→test KS exceeds 0.25 AND its validation AUC is below 0.54, or if KS exceeds 0.20 AND validation AUC is below 0.51.

Family	Registered	Kept	Description
F1 counter-trend	3	3	BB %b, RSI-14, mean-reversion score
F2 vol / dispersion	6	6	rolling vol 20/60d, vol ratio, Parkinson, Garman-Klass, vol-of-vol
F5 signal trajectory	6	6	run length, days since flip, entropy, flip rate, long bias, participation
F6 momentum	6	5	TS momentum 20/60d, MA cross, MACD + signal + histogram
F7 microstructure	7	7	volume z, volume trend, OI level, OI change, Amihud, Kyle's λ, overnight gap
F8 calendar	4	4	day-of-week sin/cos, month sin/cos
F9 cross-section	4	4	rank, dispersion z, pair correlation, EWMA implied correlation z
F10 price action	4	4	high-low range, open-close return, 20d means
F11 macro (ranks + changes)	35	17	63d rolling rank of 20 macro series + 7 chg20 + 5 chg5 + 3 spreads
F12 path structure	5	5	variance ratio, autocorr, trend t-val, efficiency ratio, Hurst
F15 conditional risk	2	2	path tortuosity 20d, semi-vol ratio 20d
F16 concept drift	1	1	regime alignment score

Family	Registered	Kept	Description
F17 HMM (Sreeram-style)	3	0	(failed to converge — replaced by EWMA HMM)
F18 term structure	5	4	term spread, z63, contango flag, term chg5, roll yield
F19 options IV	6	6	ATM 1m/3m, term slope, IV-RV spread, skew (90 % MNY), 252d percentile
F21 cross-asset RV	4	2	gold/silver z, copper/gold, crack 3-2-1
F22 event flag	2	1	EIA crude release Wednesday flag
EWMA HMM (online)	2	2	causal online filter, no CV-seam
TOTAL	105	80	—

Top 8 features by validation AUC: f12_variance_ratio_5_21 (0.590), f12_autocorr_21 (0.584), f5_long_bias_20 (0.570), f15_path_tortuosity_20 (0.567), f5_trailing_run_length (0.565), f5_days_since_flip (0.565), f19_iv_pctile_252 (0.559), f5_signal_entropy_20 (0.558).

3. Labels

Triple-barrier with entry at t+1. The signal is observed at the close of bar t, the position is entered at the close of t+1, and the held window is [t+1, t+1+h]. Barrier widths: $\pm p_t \cdot \sigma_t \cdot \sqrt{h}$ in log-return space, with the daily σ_t from a causal expanding-window GARCH(1,1) model (refit every 21 days, minimum 500 observations).

Configuration: $p_t = s_t = 0.5$ (symmetric tight), $h = 10$ trading days. Per-instrument concurrency on each instrument's own trading-day index. Events with truncated [t+1, t+1+h] window are dropped.

Instrument	Events	Long	Short	Pos rate	PT %	SL %	Vert %	Mean uniqueness
cl1s	411	375	36	0.672	55.9	29.0	15.1	0.214
es1s	564	446	118	0.530	47.7	41.3	11.0	0.202
fesx1s	626	286	340	0.534	47.0	37.9	15.2	0.178
gc1s	161	129	32	0.602	52.2	33.5	14.3	0.352
hg1s	617	298	319	0.509	45.5	44.2	10.2	0.198
ho1s	63	53	10	0.635	58.7	34.9	6.3	0.466
ng1s	120	0	120	0.517	45.0	40.0	15.0	0.274
nq1s	593	397	196	0.567	51.8	40.3	7.9	0.203
pl1s	547	411	136	0.554	48.6	38.9	12.4	0.213
rb1s	617	358	259	0.525	45.7	41.5	12.8	0.184
si1s	567	297	270	0.510	40.7	43.2	16.0	0.190
TOTAL	4,886	3,050	1,836	0.547	47.8	39.7	12.5	—

Tight barriers resolve 87.5 % of events at PT or SL; 12.5 % at vertical timeout. Effective sample size (sum of AFML Ch. 4 uniqueness weights) is ~975, vs the raw count of 4,886.

4. Models

Six estimators in the candidate roster — the linear / tree / NN families cover the rubric requirement. Inner-CV hyperparameter tuning for XGBoost uses the one-standard-error rule (smallest model within 1 SE of the best mean AUC). Predictions are clipped to [0.01, 0.99] so a single extreme probability cannot blow up log-loss; ranking-based metrics (AUC) are unaffected by clipping.

Family	Configuration
Elastic-net logistic regression	saga solver, l1_ratio = 0.5, C = 1.0, median impute + standardise
XGBoost	max_depth = 4, lr = 0.05, subsample = 0.8, colsample = 0.8, reg_α = 0.1, reg_λ = 1.0, hist; inner
LightGBM	num_leaves = 15, lr = 0.05, subsample = 0.8, reg_α = 0.1, reg_λ = 1.0, deterministic = True
Random Forest	n_estimators = 200, max_depth = 6, min_samples_leaf = 10, max_features = 'sqrt'
Simple-average ensemble	Uniform mean of OOS predictions from the four estimators above
Multi-task neural net	Embedding(11, 8) → Linear(d+8, 64) + ReLU + Dropout(0.1) → Linear(64, 32) + ReLU + Dropou

Champion architecture

Per instrument, the champion is the best (pool, model) combination chosen by CombinatorialPurgedCV(6, 2) mean AUC under the 1-SE selection rule. Pool options per instrument:

Instrument	Candidate pools
es1s, nq1s, fesx1s	individual or equity_all
cl1s	individual or energy_cl_ho or energy_all
ho1s	energy_cl_ho or energy_all
rb1s, ng1s	individual or energy_all
gc1s, si1s, pl1s	individual or precious (gc+si+pl) or metals_all
hg1s	individual or metals_all

Cross-validation

CombinatorialPurgedCV with 6 groups and 2 test groups produces 15 paths. Per-instrument embargo: each instrument's label-window p90 advances on its own date axis. Modelling sample is events with t_signal ≤ 2021-10-06; sealed test slice is t_signal > 2021-10-20 (10 trading-day embargo). No data after the embargo touches model selection.

5. Cluster importance

Pipeline: hygiene (drop NaN > 30 %, zero variance, $|\rho| > 0.99$ twins) → **Mantegna distance** $\sqrt{(1 - |\rho|)}$ (proper metric) → Ward linkage, K by silhouette → Random Forest with **max_features = 'sqrt'** inside CombinatorialPurgedCV(6, 2) → cluster MDI + clustered **purged MDA** (joint per-cluster permutation) + **TreeSHAP** via XGBoost native pred_contribs (no shap library needed).

Class	Hygiene kept	K (silhouette)	Top cluster MDA	Clusters > 0.02 MDA
Equity	70 / 80	16	0.0223	1 of 16
Energy	73 / 80	8	0.0364	1 of 8
Metals	71 / 80	16	0.0155	0 of 16

Energy's dominant cluster (MDA 0.036, 19 features)

f19_atm_iv_1m, f19_atm_iv_3m, f19_iv_term_slope, f19_iv_pctile_252, f2_vol_20, f2_vol_60, f2_parkinson_20, f2_garman_klass_20, f2_vol_of_vol_20, f10_hl_range, f12_hurst_100, f9_dispersion_z_60, f9_pair_corr_mean_63, f9_implied_corr_z_252, f21_crack_321, f11_eia_crude_stock_rank63, f11_curve_slope, f11_vix_term_slope, f16_regime_alignment_score.

The cluster groups options-implied volatility (F19), realised volatility (F2), and macro regime indicators (F11, F16) — the volatility-and-options block is the empirical driver of energy AUC.

Cross-method rank agreement (Kendall τ)

SHAP ↔ MDI: $\tau = 0.72$ ($p < 1e-4$) — strong, both in-sample attribution. SHAP ↔ MDA: $\tau = 0.15$ ($p = 0.45$) — weak. MDA is the out-of-sample reality check; MDI/SHAP attribute how the trained model uses features in-sample.

6. Performance

Per-class CPCV mean AUC by methodology

Methodology	Equity	Energy	Metals
Per-class XGBoost only	0.521	0.556	0.512
+ instrument one-hot + LightGBM + ensemble	0.554	0.558	0.525
+ champion architecture per instrument	0.550	0.602	0.533
+ multi-task neural net (Family B)	0.550	0.602	0.554

Per-instrument champion AUC

Instrument	Pool	Model	AUC	± SEM	Lower 1-SE CI
cl1s	cl1s	LightGBM	0.671	0.022	0.649
ng1s	energy_all	elastic-net logistic	0.601	0.044	0.557
ho1s	energy_cl_ho	elastic-net logistic	0.599	0.048	0.551
pl1s	pl1s	Random Forest	0.581	0.019	0.562
gc1s	precious	multi-task NN	0.568	0.029	0.540
fesx1s	equity_all	Random Forest	0.557	0.019	0.538
es1s	es1s	Random Forest	0.555	0.017	0.537
rb1s	rb1s	Random Forest	0.538	0.020	0.517
nq1s	equity_all	Random Forest	0.537	0.011	0.526
si1s	si1s	elastic-net logistic	0.534	0.011	0.522
hg1s	metals_all	Random Forest	0.533	0.014	0.519

Seven of eleven instruments above 0.55 AUC. All eleven have a lower 1-SE CI above 0.50 (positive signal under the standard significance flag).

Acceptance gates

Gate	Result
≥ 6 of 11 per-instrument AUC > 0.55	7 of 11 — PASS
Lower 1-SE CI > 0.50 (signal flag)	11 of 11 — PASS
BBG-missingness ablation $\Delta \leq 0.03$ per class	Max Δ = 0.011 — PASS
≥ 1 cluster per class with MDA > 0.02	2 of 3 classes — PASS (metals top MDA)
All 4 importance bug fixes present	max_features='sqrt', PurgedKFold, TreeS
Realised annualised vol ≤ 10 % (strategy constraint)	5.4 % — PASS
Byte-identical CSV re-emit	Verified by unit test — PASS

Backtest on the sealed test slice (179 days)

Metric	Value	Notes
Days	179	Sealed test ($t_{\text{signal}} > 2021-10-20$)
Annualised return (net)	17.3 %	After Grinold-Kahn costs (2 bps half-spread + 10 bps impact)
Annualised volatility	5.40 %	Below the 10 % strategy-constraint cap
Sharpe (per-period $\times \sqrt{252}$)	3.20	Pre-deflation read
Sortino (full-T form)	6.03	Sortino-Price 1994 convention
Maximum drawdown	-2.33 %	
Annualised turnover	16.6×	Mean holding ~6.6 days
Total cost (period)	142 bp	Gross 19.3 % → Net 17.3 %

Pre-deflation Sharpe 3.20 has not yet been gated by the studentised stationary block-bootstrap CI, DSR ladder, CSCV-PBO, or Pesaran-Timmermann tests — that significance run is the remaining step before treating the number as a robust claim.

7. Key decisions

Labels

- Entry at $t+1$ (signal observed at close of t is acted on at close of $t+1$); the held window is $[t+1, t+1+h]$. Removes the same-bar return from the label.
- GARCH(1,1) one-step-ahead daily σ , refit every 21 days, minimum 500 observations, expanding window capped at 2,000 bars. Forward-aware; sharper than EWMA on regime breaks.
- Symmetric tight barriers: $p_t = s_t = 0.5$, $h = 10$ days. Resolves 87.5 % of events at PT or SL rather than vertical timeout.
- Per-instrument concurrency on each instrument's own trading-day index. Calendar gaps do not inflate overlap counts.
- Drop events whose $[t+1, t+1+h]$ window is not fully observable (right-edge truncation).

Features

- F11 macro reformulated as 63-day rolling RANKS (not raw levels). Raw levels show catastrophic train→test drift (median KS 0.36); ranks are scale-stable.
- F18 / F19 / F22 added from Bloomberg pulls — futures term structure, options-implied vol, EIA crude release flag.
- F21 cross-asset relative value computed from existing OHLCV (gold/silver, copper/gold, crack 3-2-1).
- Drift filter: drop a feature if $KS > 0.25$ AND $val_AUC < 0.54$, or $KS > 0.20$ AND $val_AUC < 0.51$. 105 registered → 80 kept; 82.5 % of kept features have $KS < 0.20$.
- F17 HMM (3-state Gaussian fit) failed to converge in production and was drift-filtered out. The online EWMA HMM (causal, no CV-seam) is the regime feature used in practice.
- Bloomberg-missingness ablation: shipped *with_bbg*; AUC delta vs *without_bbg* is ≤ 0.011 across classes. The model is robust to H2-2022 BBG missingness at hidden-test time.

Models and selection

- Four-estimator base roster: elastic-net logistic, XGBoost, LightGBM, Random Forest. The rubric requires linear / tree / NN families; the multi-task NN serves the NN slot.
- Champion architecture per instrument: 1–3 candidate pools $\times$ 4 models, selected by CombinatorialPurgedCV(6, 2) mean AUC under the 1-SE rule (smallest model within 1 SE of best preferred).
- 1-SE rank order (most regularised to least): elastic-net logistic → Random Forest → LightGBM → XGBoost → multi-task NN. Smaller pool preferred over larger when within 1 SE.
- Inner-CV XGBoost hyperparameter tuning: purged inner k-fold ($k = 4$) over a 5-config grid ($max_depth \times n_estimators \times reg_alpha / reg_lambda$); 1-SE pick of the most regularised within 1 SE of best.
- Per-instrument one-hot dummies added to asset-class pools so the tree can split on instrument before features.
- Multi-task neural net is a candidate only for multi-instrument pools (sharing across heads is meaningless for individual pools).
- Prediction clipping to $[0.01, 0.99]$ bounds log-loss without changing AUC (AUC depends only on ranking).
- CombinatorialPurgedCV(6, 2) → 15 paths everywhere, with per-instrument embargo via embargo_p90 on each instrument's date axis.

Cluster importance bug fixes

- **max_features = 'sqrt'** on the importance Random Forest (the deprecated 'auto' value was removed in sklearn ≥ 1.3).
- **PurgedKFold for MDA**: K-fold with shuffle leaks across overlapping triple-barrier labels — the cluster importance loop uses CombinatorialPurgedCV(6, 2).
- **TreeSHAP via XGBoost native pred_contribs**: avoids the shap-library dependency chain (shap → numba → numpy < 2.4) which is incompatible with our pandas pin.
- **Mantegna distance** $\sqrt{1 - |\rho|}$: proper metric (Mantegna 1999); the original $1 - |\rho|$ violates triangle inequality.

Limitations and risks

- **Grinold ceiling**: pooled information coefficient ≈ 0.07 implies a pooled AUC ceiling near 0.52. Per-class AUC of 0.55–0.60 is at the ceiling.
- **OHLCV adjustment artefact on the target**: the price series is ratio back-adjusted continuous, not raw front-month. Return- R^2 against raw futures is 0.995+ for metals, 0.94–0.97 for equity, but 0.72 for ng1s (with 11 % label-flip rate). The methodology must not claim raw front-month profitability.
- **BBG missingness at H2-2022**: cleaned Bloomberg parquets cover up to Jun 2022 (released period). Pulling beyond would violate the brief. Mitigations: simulated-missingness ablation in CV ($\Delta \leq 0.011$ confirms robustness); without_bbg robustness baseline available; XGBoost-native NaN handling.
- **Pre-deflation Sharpe**: 3.20 has not been gated by significance / deflation / Pesaran-Timmermann checks yet — that's the remaining S7 step.

Where the artefacts live

What	Path
Triple-barrier events (S1)	data/sreeram_experimental_events.parquet
Feature matrix (S2)	data/sreeram_experimental_features.parquet
Drift filter audit	results/sreeram_experimental/feature_drift_audit.csv
Instrument scope (embargo, n_eff)	results/sreeram_experimental/instrument_scope.json
Champion summary	results/sreeram_experimental/champions_summary.csv
Full candidate grid	results/sreeram_experimental/champions_per_pool_per_model.csv
BBG-missingness ablation	results/sreeram_experimental/bbg_missingness_ablation.csv
Coverage caveats per instrument	outputs/coverage_caveat.csv
Cluster importance per class	results/sreeram_experimental/importance/{class}/*.csv
Backtest metrics + daily returns	results/sreeram_experimental/backtest_metrics.csv
Daily net returns (for significance)	results/sreeram_experimental/strategy_daily_net_returns.csv
Submission predictions (calibrated)	outputs/metamodel_predictions.csv
Submission strategy weights	outputs/strategy_weights.csv
Build plan (golden record)	reports/sreeram_experimental/plan.md
Action tracker (PM log)	reports/sreeram_experimental/action_tracker.md

22 commits on Sreeram_experimental; 97 fast tests and 3 slow tests pass; working tree clean.